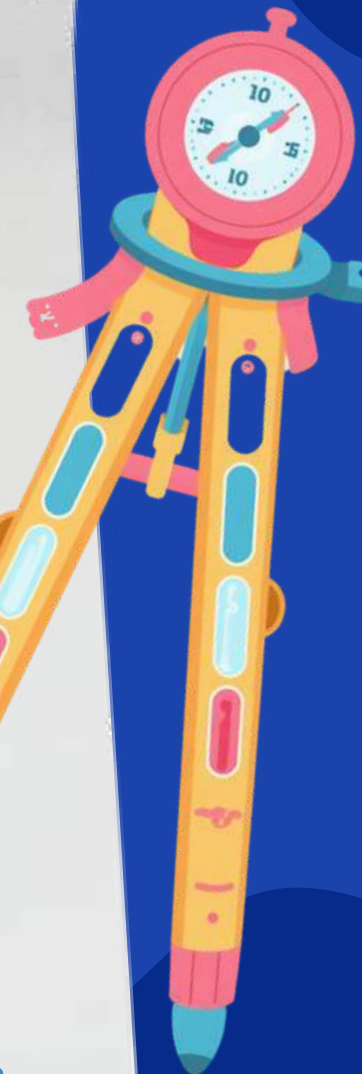




$\frac{1}{2}$

**= GROUP IT,
MAP IT!**

×





Ready for the Picnic!



Our Story Begins...

Ananya and Kabir are packing for their school picnic to Kanha National Park. They have lots of snacks, games, and supplies — but everything is jumbled up! Can you help them sort and group their things? Let's get packing! 🎒





Let's Sort Indian Street Food!

Help Ananya and Kabir sort their favourite Indian street foods!
Samosa, Murukku, Pani Puri, Banana Chips, Pakora, Mirchi Bajji,
Kolhapuri Misal

Think & Sort!

Which snacks are CRISPY? Which are SPICY? Can some be both?
Pick each food and decide — sort them into groups!





What is a Set?



A set is a group of well-defined objects that belong together based on a common rule or property.

Key Definition

Example 1: Rohan's Geometry Box — {Triangle, Circle, Square, Rectangle, Ruler}

Example 2: Traditional Indian Breakfast — {Idli, Dosa, Poha, Paratha, Upma}
Each group has a clear rule: you know exactly what belongs in it!



Single Sets in Action!

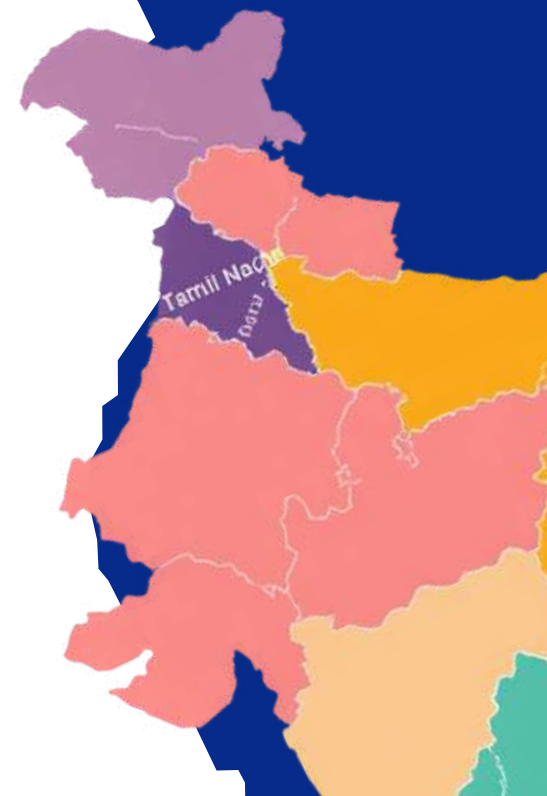
A set is a collection of well-defined objects. Here is one set: South Indian States!

This set contains exactly 5 states that belong together in one group.

Set: South Indian States

{ Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, Telangana }

All 5 states belong to the same set — South India! Can you think of another set?

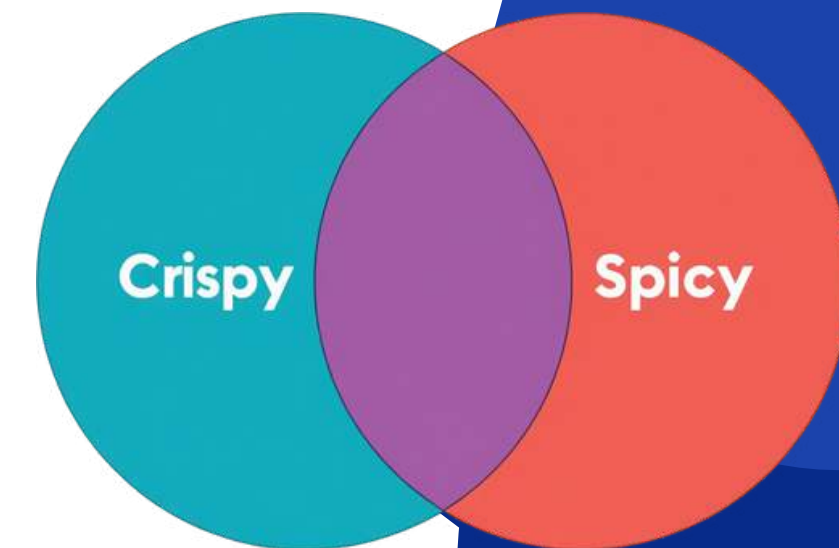


Tamil Nadu,
Karnataka

!lɹɔ cldy
Karnataka



Overlapping Sets! Crispy & Spicy Snacks



Crispy Snacks ●

Samosa, Murukku, Banana Chips,
Kolhapuri Misal

These snacks are crunchy and
crispy to eat!



Spicy Snacks ●

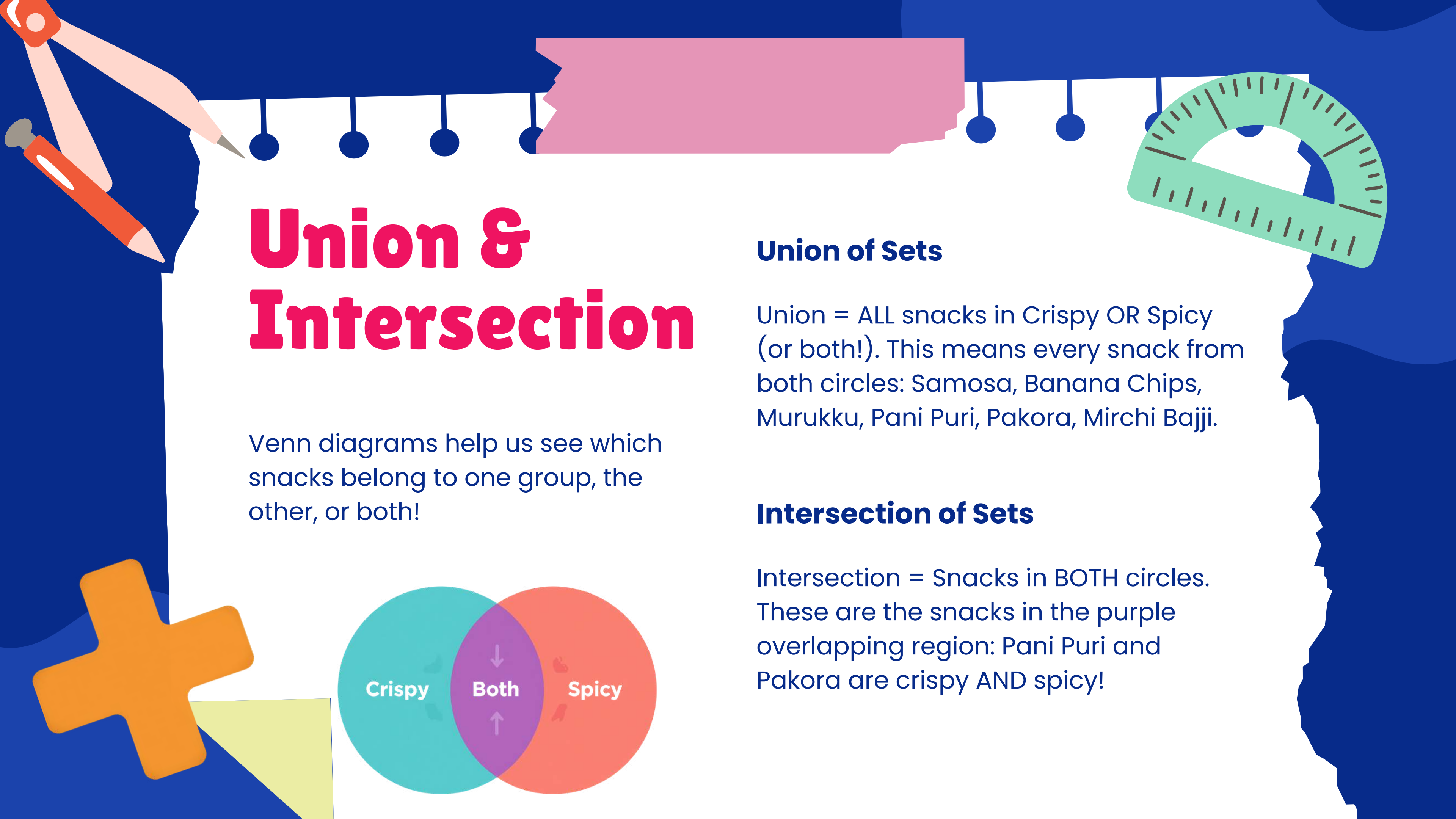
Mirchi Bajji, Kolhapuri Misal

● Both Crispy & Spicy:

Pani Puri, Pakora

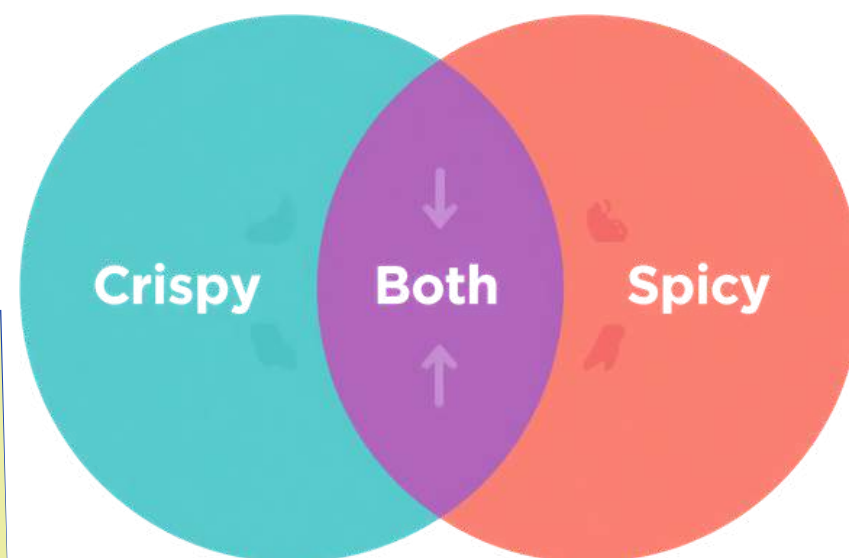
Some snacks belong to BOTH
groups!





Union & Intersection

Venn diagrams help us see which snacks belong to one group, the other, or both!



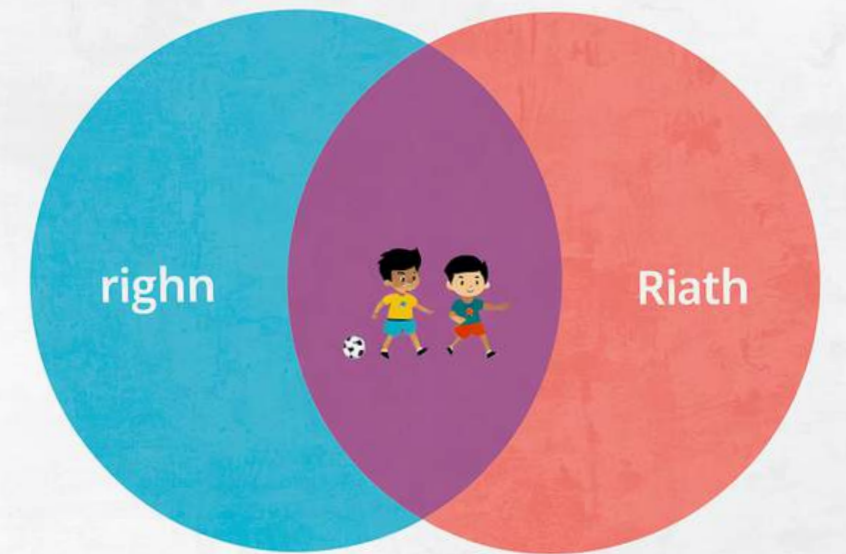
Union of Sets

Union = ALL snacks in Crispy OR Spicy (or both!). This means every snack from both circles: Samosa, Banana Chips, Murukku, Pani Puri, Pakora, Mirchi Bajji.

Intersection of Sets

Intersection = Snacks in BOTH circles. These are the snacks in the purple overlapping region: Pani Puri and Pakora are crispy AND spicy!

Sports Time!



Who plays cricket? Who plays badminton? Who plays both? Let's sort the colony kids into sets!

Only Cricket



Arjun
Rohan
Priya
Deepak

(These kids play only cricket — not badminton!)

Only Badminton

Sneha
Meera
Karthik
Anusha

(These kids play only badminton — not cricket!)



Disjoint Sets

Railway vs Flying Vehicles

Some sets have NO common elements — they don't overlap at all!
These are called Disjoint Sets.

 Railway Vehicles: Train, Metro, Tram, Monorail

 Flying Vehicles: Aeroplane, Helicopter, Hot Air Balloon

No vehicle can be both a railway vehicle AND a flying vehicle — so these two circles never overlap!



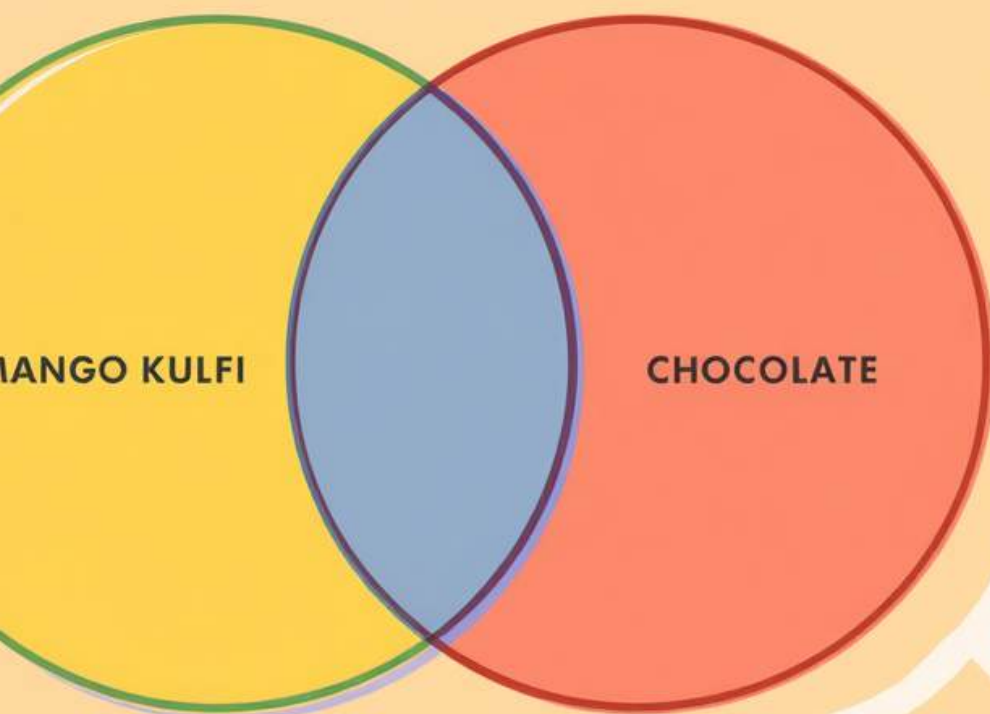


The Universal Set!

All 30 students in Class 5A love ice cream! The Universal Set includes everyone. Inside, some love Mango Kulfi and some love Chocolate Ice Cream — and some love both!

Class 5A Ice Cream Lovers

Universal Set = All 30 students in Class 5A
Mango Kulfi lovers and Chocolate lovers
are subsets inside the big group!
Overlapping region = Students who love
BOTH flavours!





Problem Time!

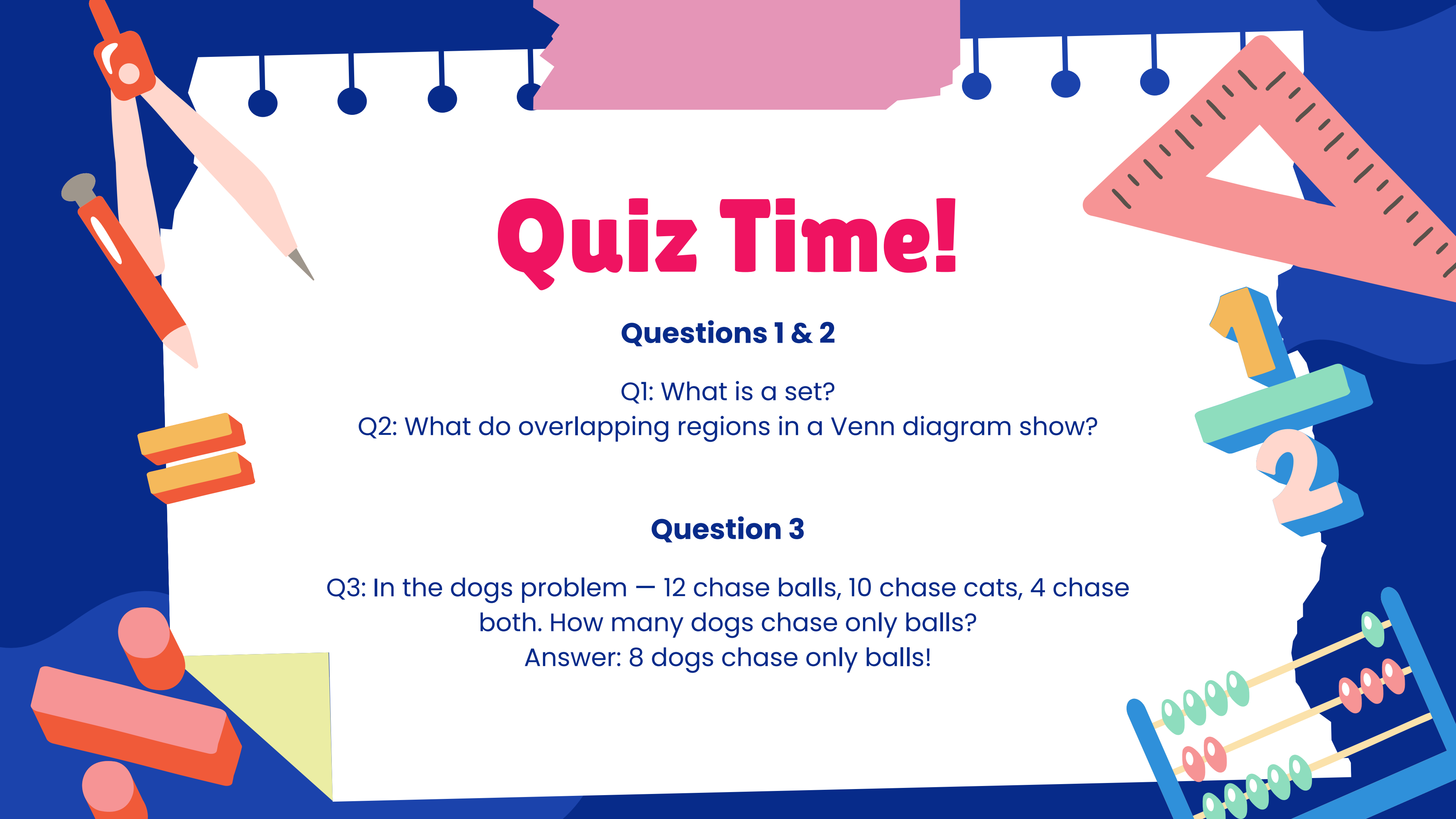
Dogs Chasing Balls & Cats

12 dogs chase balls, 10 dogs chase cats, and 4 dogs chase both.
How many dogs chase only balls?

Balls Only: $12 - 4 = 8$ dogs Both: 4 dogs Cats Only: $10 - 4 = 6$ dogs

✓ Answer: 8 dogs chase only balls!





Quiz Time!

Questions 1 & 2

Q1: What is a set?

Q2: What do overlapping regions in a Venn diagram show?

Question 3

Q3: In the dogs problem — 12 chase balls, 10 chase cats, 4 chase both. How many dogs chase only balls?

Answer: 8 dogs chase only balls!

THANK YOU!

